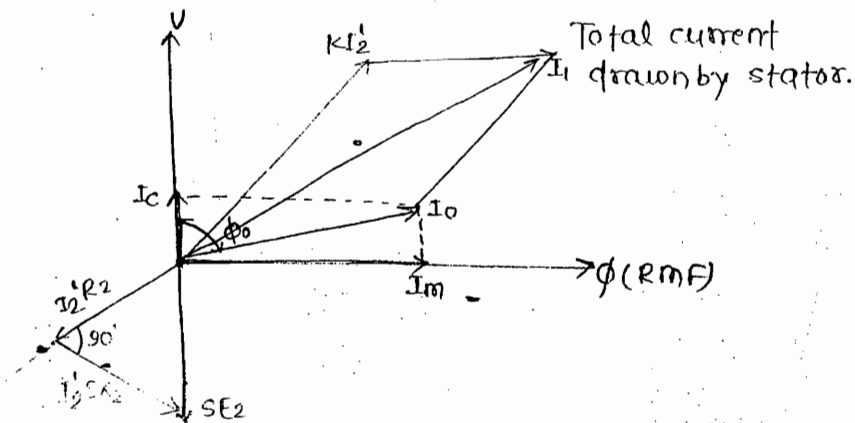


$$SE_2 - I_2'(R_2 + jX_2) = 0 \quad (V)$$

$$I_2' = \frac{SE_2}{R_2 + jX_2}$$



- * On load rotor current increase proportionally to produce its own flux in the air gap which is stationary w.r.t stator RMF. but opposes it acc. to Lenz law & reduce it.
- * Therefore stator draws more current proportionally to maintain the flux const in the air gap.
- * Whenever rotor is loaded it will immediately call current from stator. ($K I_2'$)
- With load PF improves but operate at low PF comparative to an eq. Xmer.

Transformer

- (1) Ele. to ele. energy conversion through magnetic medium in high permeability core
- (2) Require less magnetising current
- (3) NL current 2-6% of rated
- (4) Low lagging PF on NL
- (5) PF improves with load.
- (6) Concentrated wdg.

Induction motor --

- Ele. to mech. energy conversion through magnetic medium in the air gap of high reluctance.
- Requires large magnetising current.
 - Around 30-40% of rated.
 - Very much low lagging PF.
 - PF improves with load. but compare low.
 - Distributed wdg produce RMF.

(7.) Contain Core loss & Cu loss only

(8.) Induced emfs depends on turns ratio.

(9.) Freq. of 1° , 2° same

(10.) 2° is never SC

(11) Ele. loading across 2° terminals

(12) leakage flux less & leakage reactances

(13) Constant flux device

(14) Mutual indⁿ

Stator core loss, no loss in air gap & rotor core loss negligible & includes mech. loss. with Cu losses of stator & rotor.

Induced emfs depends on turns ratio only at standstill but vary with slip under running.

freq. of stator, rotor same only at standstill but vary with slip.

But rotor should be essentially SC.

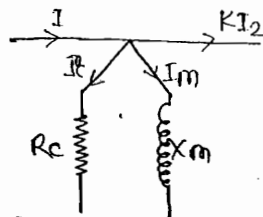
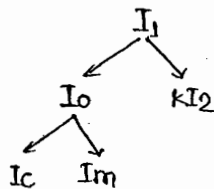
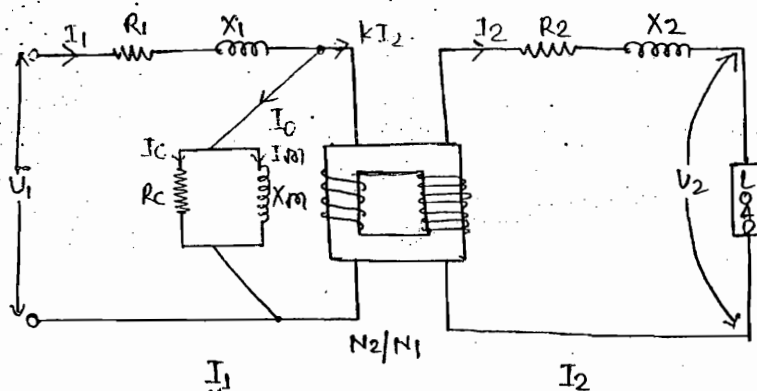
mech. loading across rotor shaft.

Leakage flux high & leakage reactance.

Constant flux m/c.

Mutual indⁿ

* T/F eq. Ckt Representation $\rightarrow$



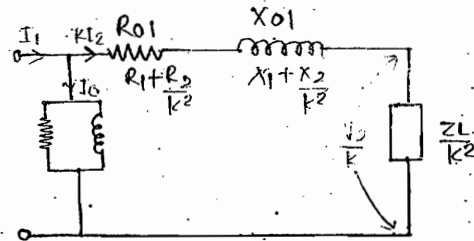
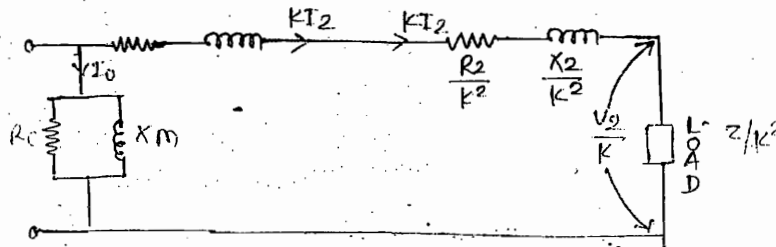
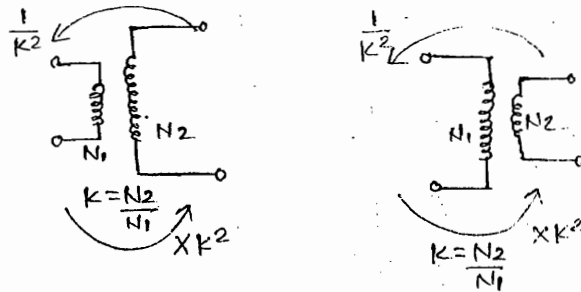
R_c, X_m depends on I_c & I_m .

$$R_c = \frac{V_1}{I_c} \gg X_m = \frac{V_1}{I_m} \quad \left[\text{fictitious / Imag. value} \right]$$

$$I_c < I_m$$

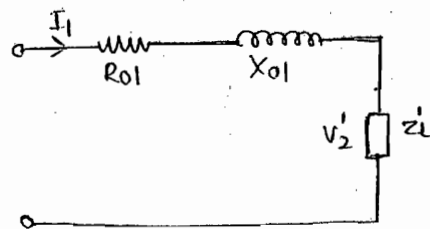
$$I_1^2 R_2' \Rightarrow I_2^2 R_2$$

$$R_2' \Rightarrow \left(\frac{I_2}{I_1}\right)^2 R_2$$



approx. eq. ckt Neglecting shunt branch

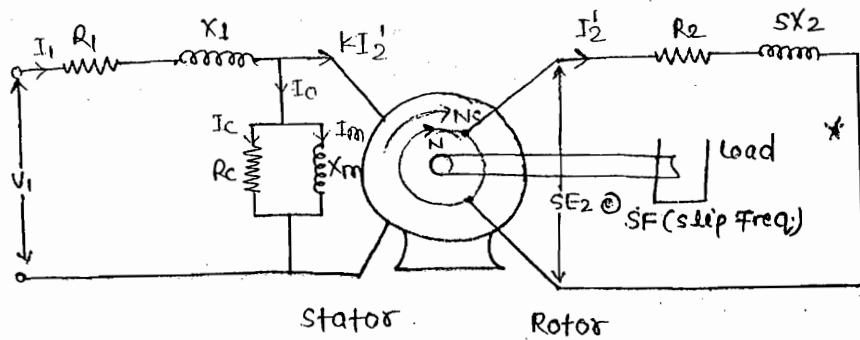
Ref to 1°



Note → As the principle of operation behaviour is identical to both TF & Im which is also called as rotating TF. The TF eq. ckt is used for representing Im. For analysis purpose.

* However there are 2 major diff which need to be resolved:-

- (i) freq. of rotor is slip freq.
- (ii) Loading is mech. across the shaft.



S_{RMF} w.r.t stator @ N_s

R_{RMF} w.r.t stator @ $S N_s + N$

∴ Rotor itself rotate @ N in the same dirn of mag. Field

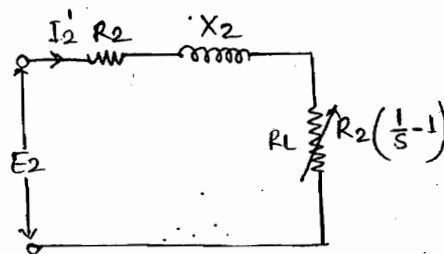
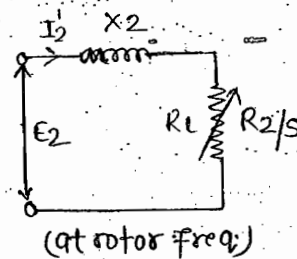
$$N_s - N + N \\ = N_s$$

Note:- Both stator RMF & rotor RMF runs at syn. speed in the air-gap & make the rotor to rotate at any speed N .

* The absolute flux in space will always rotate at syn. speed w.r.t a stationary point in space of stator but w.r.t rotor it is at $S N_s$.

$$I_2' = \frac{SE_2}{R_2 + jX_2S}$$

$$I_2' = \frac{E_2}{\frac{R_2}{S} + j\frac{X_2}{S}} \Rightarrow$$



* If $I_m \rightarrow$ open (NL)

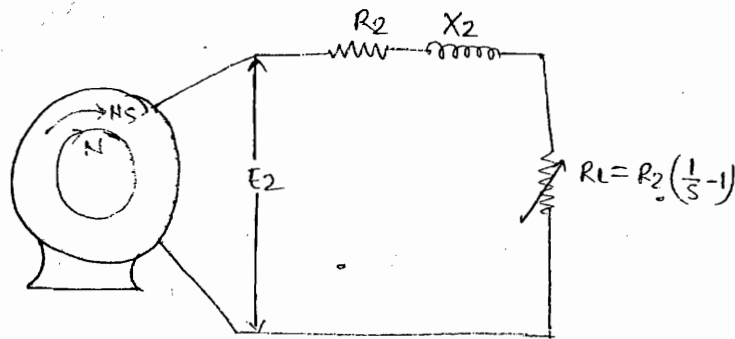
$$S \rightarrow 0$$

$R_L \rightarrow \infty$ (OC) like TF

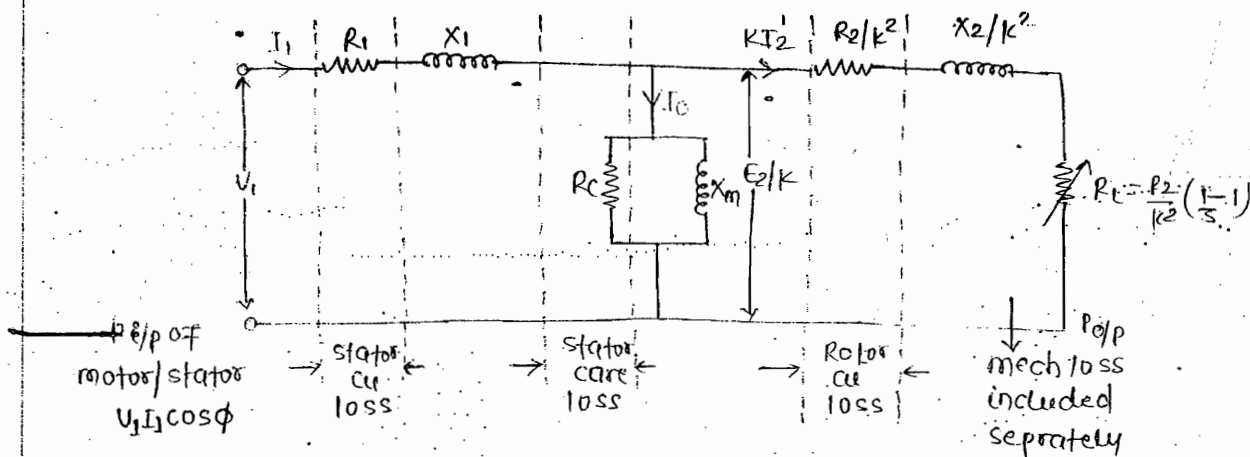
* $I_m \rightarrow$ Rated load

$$S \rightarrow 1$$

$R_L \rightarrow 0$ (SC)

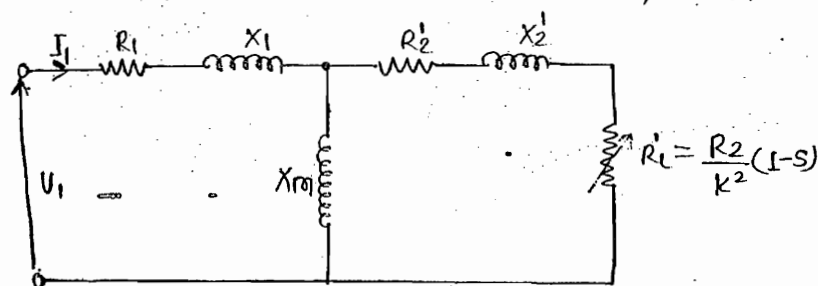


$$k = \frac{\text{Rotor turns/phase}}{\text{Stator turns/phase}} = \frac{E_2 \text{ @ stand still}}{E_1}$$

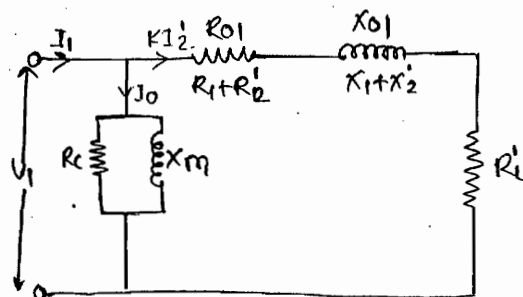


* IEEE standard ckt representation → Shunt branch resistance is eliminated in the ^{in the} std ckt but core losses & mech. losses should be

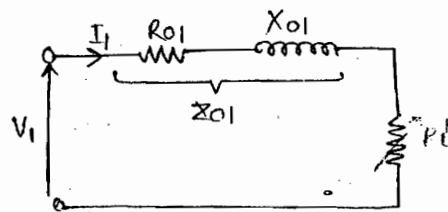
included in net power caln. This will be accurate analysis.



* Approx Eq. ckt →



Neglecting shunt branch $\rightarrow$

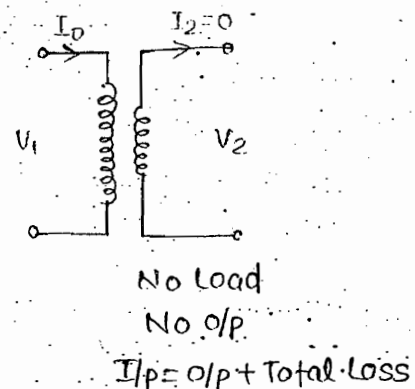
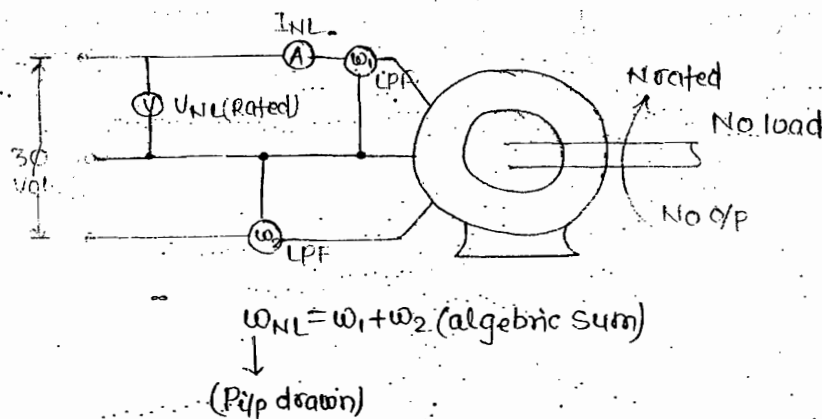


DATE-07/12/14

* Determination of eq. ckt parameters $\rightarrow$ It requires 3 tests to be performed:-

- (1.) No Load
- (2.) Blocked rotor test
- (3.) Dc test (or) Voltmeter-Ammeter method.

(1.) No Load test $\rightarrow$



* Apply rated vol. across the stator & allow it to run freely on NL.

* Connect necessary meters & essentially LPF wattmeters are req.

* The 3 ϕ power is measured using 2 wattmeter method & the total power is algebraic sum of both meter.

* On the NL as o/p is 0 the i/p power drawn is to supply all the losses at NL. Therefore the wattmeter reading is concerned as loss.

Total loss = Stator cu loss ✓

Stator core loss ✓

Rotor core loss $\rightarrow$ Neglected due to slip freq. Less

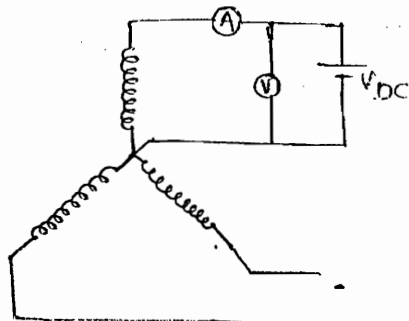
Rotor cu loss $\rightarrow$ Neglected

mechanical loss ✓

* On NL the NL power drawn is considered to supply stator core.

* In order to cal. rotational or constant losses stator resistance need to be calculated to eliminate stator Cu loss.

3.) DC test (OR) V-m, A-m method →



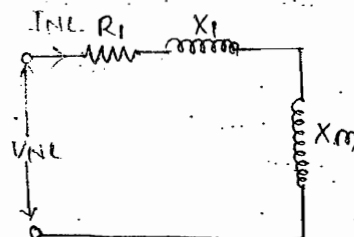
V/A Ratio R_{DC}/phase

$$R_{AC}/\text{ph.} = \underbrace{1.2-1.6}_{\substack{\text{skin effect} \\ \& \\ \text{Hot resistance}}} R_{DC}/\text{ph}$$

$$W_{NL} - 3I_{NL}^2 R_1 = \text{Constant Loss/Rotational loss}$$

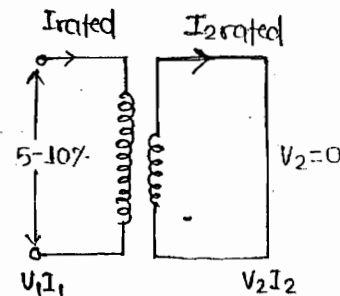
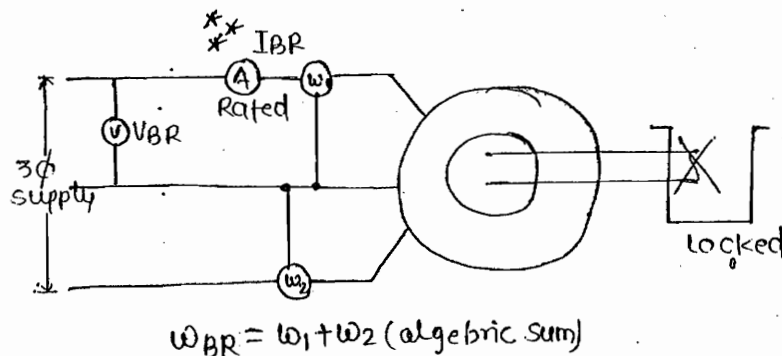
$$Z_{NL} = V_{NL}/I_{NL}$$

$$Z_{NL} = R_1 + j(X_m + X_l)$$



From the meter reading Z_{NL} is calculated, from blocked rotor test X_l is cal. & the magnetising reactance X_m is determined using the above eqⁿ.

2.) Blocked Rotor test → The rotor is blocked initially to ensure no rotation. vol. is applied across the stator carefully using the suitable auto Xmer to ensure rated current.

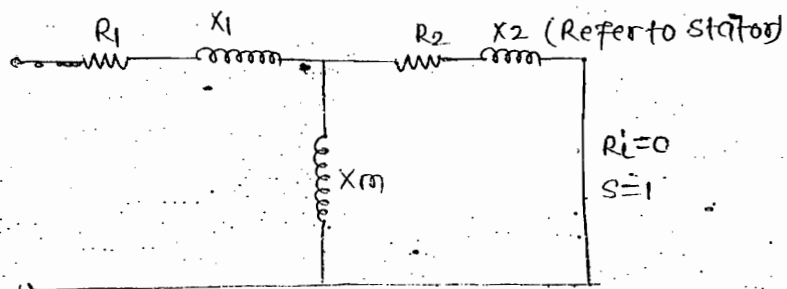


$$P = \frac{2\pi NT}{60} \quad (N=0)$$

$P=0$ [so we have right to take w-m reading as loss]

So w-m readings are $\rightarrow$ Total losses

- $\rightarrow$ Stator core loss $\times$ (Vol. is low)
- $\rightarrow$ St. Cu loss $\checkmark$ (Rated current)
- $\rightarrow$ Rotor core loss $\times$ (ϕ ↓, V ↓)
- $\rightarrow$ Rotor Cu loss $\checkmark$
- $\rightarrow$ Mech. loss = 0 (Not running of motor)



Eq. ckt on blocked rotor condⁿ.

* This tests gives the information of Cu loss at rated current condⁿ?

$$\frac{V_{BR}}{I_{BR}} = Z_{BR} = R_{BR} + jX_{BR}$$

$$Z_{BR} = [R_1 + jX_1] + jX_m \parallel (R_2 + jX_2)$$

$$Z_{BR} = (R_1 + jX_1) + \frac{jX_m (R_2 + jX_2)}{j(X_m + X_2) + R_2} \times \frac{[R_2 - j(X_m + X_2)]}{[R_2 - j(X_m + X_2)]}$$

$$= (R_1 + jX_1) + \frac{jX_m [R_2^2 - jR_2(X_2 + X_m) + jR_2X_2 + X_2(X_2 + X_m)]}{R_2^2 + (X_2 + X_m)^2}$$

$$= (R_1 + jX_1) + \frac{jX_m [R_2^2 - jR_2X_m + X_2(X_2 + X_m)]}{R_2^2 + (X_2 + X_m)^2}$$

Equating imaginary terms

$$X_{BR} = X_1 + \frac{X_m [R_2^2 + X_2(X_2 + X_m)]}{R_2^2 + (X_2 + X_m)^2}$$

X_m is practically high compared R_2
 $(X_2 + X_m)$ is also high. let it be (X_{2m})

$$X_{BR} = X_1 + \frac{X_m [R_2^2 + X_2 \cdot X_{2m}]}{R_2^2 + X_{2m}^2}$$

$$= X_1 + \frac{X_m [R_2^2 / X_{2m} + X_2]}{[R_2^2 / X_{2m} + X_{2m}]}$$

* $\frac{R_2^2}{X_{2m}}$ is practically very low

$$X_{BR} = \frac{X_1 + X_m X_2}{X_{2m}} = \frac{X_1 + X_m X_2}{X_2 + X_m}$$

$$X_{BR} = \frac{X_1 + \frac{X_m X_2}{X_m}}{\frac{X_2 + X_m}{X_m}}$$

$$= X_1 + \frac{X_2}{\left[\frac{X_2}{X_m} + 1 \right]}$$

→ practically low, negligible

$$\boxed{X_{BR} = X_1 + X_2'}$$

$X_2 \rightarrow X_2'$ (Because of refer to stator)

* There is no practical method to separate or measure X_1, X_2' .

Therefore it is approximated as

$$\boxed{X_1 \approx X_2' \approx \frac{X_{BR}}{2}}$$

Equating the real parts of eqⁿ

$$R_{BR} = R_1 + \frac{R_2 X_m^2}{R_2^2 + X_{2m}^2}$$

$$= R_1 + \frac{R_2}{\left(\frac{R_2^2}{X_m^2} + \frac{X_{2m}^2}{X_m^2} \right)}$$

→ low

$$R_{BR} = R_1 + \frac{R_2}{(X_{2m}/X_m)^2}$$

$$R_2 = (R_{BR} - R_1) \left(\frac{X_{m2}}{X_{m1}} \right)^2$$

$$R_2' = (R_{BR} - R_1) \left(\frac{X_{m2}}{X_m} \right)^2$$

From Blocked rotor test X_1, X_2', R_2' & Cu losses are calculated.

From both the tests all the eq. ckt parameters are evaluated including η

$$\eta = \frac{\text{o/p}}{\text{o/p} + \text{const loss} + \text{variable loss}}$$

$\downarrow$ $\downarrow$
 Iron Cu
 Fric. loss
 Windage

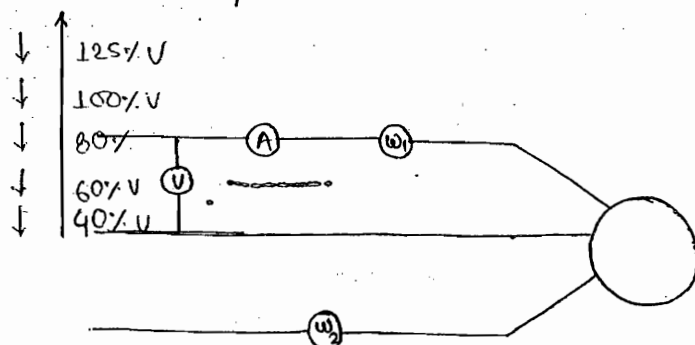
* Separation of rotational losses $\rightarrow$ * A NL test is performed to separate iron & mech. losses. using a suitable auto Xmes apply different vol. across the stator. from 125% rated to around 40% rated.

* As the motor is running on NL the torque requirement is very low & the variation of slip with variation in V is low upto a breakdown point.

Therefore in the testing condⁿ the speed of motor is approx. same (variation is less)

$$T \propto sV^2 \quad (\because E \propto V)$$

* Iron losses $\propto \phi$, mech. losses $\propto N$



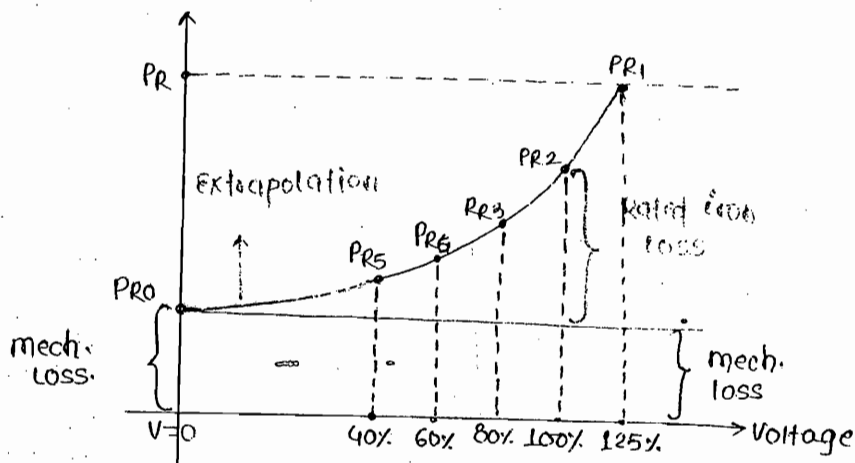
$$(1) \quad V_{NL1} \quad I_{NL1} \quad W_{NL1} \rightarrow W_{NL1} - 3I_{NL1}^2 R_1 = P_{R1}$$

$$(2) \quad V_{NL2} \quad I_{NL2} \quad W_{NL2} \rightarrow W_{NL2} - 3I_{NL2}^2 R_1 = P_{R2}$$

$$(3) \quad 80\% \quad W_{NL3} - 3I_{NL3}^2 R_1 = P_{R3}$$

$$(4) \quad 60\% \quad W_{NL4} - 3I_{NL4}^2 R_1 = P_{R4}$$

$$(5) \quad 40\% \quad W_{NL5} - 3I_{NL5}^2 R_1 = P_{R5}$$



(Curve is parabolic along with V^2)

* Circle diagram →

- * It is the graphical representation of approximate eq. ckt.
- * Once a circle dia. is plotted any analysis parameter can be simply & rapidly determined (approximately) w/o approaching the motor.
- * The locus of current in a series RL ckt will make a circle with a dia. V/X .
- * If a chord in a circle is bisected it passes through its center.
- * In order to plot it is requires data from NL test, blocked rotor test & the dc test.
- * Circle dia. is plotted on a vertical vol. as ref. axis. Therefore blocked rotor test values need to be adjusted to rated vol. base. for plotting circle dia.

No load test → V_{rated}

$$V_{NL} \quad I_{NL} \quad W_{NL} = \sqrt{3} V_{NL} I_{NL} \cos \phi_{NL}$$

$$* V_{rated} \leftarrow I_{NL} \quad \phi_{NL} = \cos^{-1} \left[\frac{W_{NL}}{\sqrt{3} V_{NL} I_{NL}} \right]$$

Blocked Rotor → (5-10% V_{rated}) → V_{BR}

I_{BR}
 W_{BR} } (Values adjust to V_{rated})

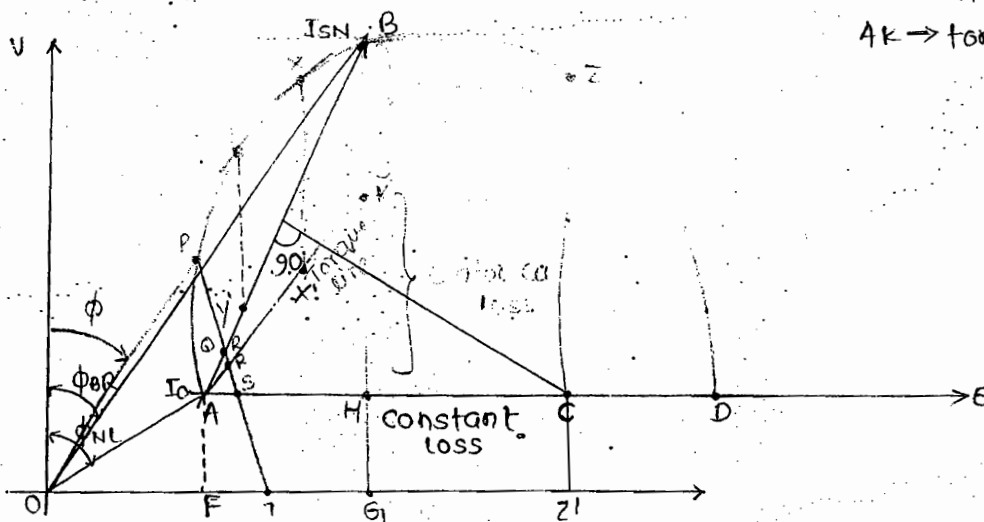
$$V_{BR} \quad I_{BR} \quad W_{BR} = \sqrt{3} V_{BR} I_{BR} \cos \phi_{BR}$$

$$\text{If } V_{rated} \quad \text{let } \phi_{BR} = \cos^{-1} \left[\frac{W_{BR}}{\sqrt{3} V_{BR} I_{BR}} \right]$$

$$I_{SN} = \frac{V_{rated}}{V_{BR}} \times I_{BR}$$

Steps →

- AB $\rightarrow$ o/p line.
AK $\rightarrow$ torque line



- * The vertical distance length of BG represents total losses for a current I_{SN} . As it is blocked rotor mode mech. losses are zero, but it gives sufficiently accurate value because rotor core losses are neglected under such condⁿ.
- * Length of BH represents variable losses i.e. stator cu loss & rotor cu loss for I_{SN} .
- * From dc test stator cu losses are cal. for a current I_{SN} value & using power scale a point K is located where length of HK represents stator cu loss.
- * Consequently the other length BK represents rotor cu loss.

power scale:- watts/cm.

$$l(BG)_{\text{cms}} = W_{SN}(\text{watts})$$

$$1\text{cm} = \frac{W_{SN}(\text{watts})}{l(BG)_{\text{cms}}}$$

Join AK, which is called as torque line

Data obtained from circle dia:-

(1) max^m torque:- Draw a tangent on circle parallel to torque line to meet at point X. Drop a vertical on to the torque line to meet

at X'.

$$l(XX') \times \text{power scale} = T_{\text{max}} \text{ in watts}$$

Synchronous watts

$$T = \frac{60}{2\pi N_s} \times R P i/p$$

$$T \propto P$$

→ Constant

(2) max^m o/p → Draw a tangent parallel to o/p line to meet at point on the circle dia. Drop a vertical on to vertical line to meet at Y.

$$l(YV) \times \text{power scale} = \text{max o/p in watts}$$

(3) Starting torque → start.

$$T = \frac{60}{2\pi N_s} \times R P i/p$$

$$= \text{Const.} \times \frac{\text{Rotor cu/loss}}{s}$$

$$T = \text{const} \times \frac{\text{Rotor cu/loss}}{1}$$

$$\boxed{T_{st} \propto \text{Rotor cu/loss}} \quad (s=1)$$

$$T_{st} \propto \text{length of (BK) in watts}$$

$$T_{st} = \frac{60}{2\pi N_s} l(BK) \quad \text{--- max power scale (N-m)}$$

(4) I/p →

$$I/p = O/p + \text{Total losses}$$

Extend point c on to the circle dia. to meet at z & the bottom axis to meet at z'.

$$l(zz') \times \text{power scale}$$

eg:- From the test data given plot circle dia. & cal. all its performance parameter when the motor operates at full load.

* Calculate the rated current of motor & locate a point p on the circle dia. so that length of op represents I_{rated} .

Draw a vertical line parallel to vol. axis to meet at point Q, R, S, T

$$l(ST) \times \text{power scale} = \text{const. loss}$$

$$l(RS) \times \text{power scale} = \text{stator cu loss}$$

$$l(RQ) \times \text{Power scale} = \text{Rotor cu loss}$$

$$I/p = O/p + \text{total losses}$$

$$PT = PQ + TQ$$

$$l(PQ) \times \text{Power scale} = O/p$$

$$l(PT) \times \text{Power scale} = I/p$$

$$l(TQ) \times \text{power scale} = \text{total loss}$$

$$\eta = \frac{O/p}{I/p} = \frac{PQ}{PT}$$

$$\text{slip} = ? , \text{ rotor cu loss} = s \cdot \text{Rotor power } I/p$$

$$l(RQ) = s \cdot l(PR)$$

$$s = \frac{RQ}{RP}$$

$$N = N_s(1-s)$$

known

$$T_g = \frac{60}{2\pi N_s} \cdot l(RP) \times \text{power scale} \quad \text{N-m}$$